Value Iteration

Lecture Notes — Reinforcement Learning 2026–27, Deck L04

Based on slides L04 (slides 1–75) and the companion note `L04-1-value-iteration.pdf`

Abstract

This lecture answers the question left open by the MDP lectures: *how do we actually compute* V^π and, above all, the optimal policy π^*? We first compute V^π exactly (a linear system) and iteratively (a fixed-point iteration), introduce *contraction mappings* as the tool that makes iteration converge, show that both Bellman backups are γ-contractions, and conclude with **value iteration**, its convergence rate, the quality of the greedy policy it returns, and the number of iterations needed for an ϵ-optimal policy.

How these notes are organised

- **Sections follow the slide deck**, with the slide range in italics next to each title. Formal material from the companion note is merged where the slides introduce the topic.
- **Numbering.** Definitions, theorems and corollaries *1–8* and algorithms *A, B* keep the numbering of the companion note, so that its self-check list and proof videos map directly (see Appendix B). Items numbered *S, L, E, F* are not in the companion note: S/L = propositions/lemmas derived on the slides (or needed for completeness), E = worked examples, F = facts recalled from the previous lecture (L03).
- **Coloured boxes.** Definitions (green), theorems/corollaries (red), propositions/lemmas (orange), examples (teal), algorithms (violet), recalled facts (grey), and blue *Intuition* boxes that explain the "why" in plain words.
- **Running example.** A tiny two-state, two-action MDP (Example E1) is used for all worked numbers and for the data-driven plots.
- **Supplementary material** not covered on the slides (the existence/rate part of Banach's theorem) is in Appendix A, clearly labelled.

Throughout, rewards lie in $[0,1]$ (so $R_{\max} = 1$), $\gamma \in [0,1)$, $\|\cdot\|_\infty$ is the sup-norm, and ln is the natural logarithm (the slides write log for it).

Contents

1 Recap: Sequential Decision Making and MDPs *(slides 2–12)* **3**

1.1 Sequential Decision Making and the MDP *(slides 3–4)* 3

1.2 Policies, Value Functions and Q-Functions *(slides 5–6)* 3

1.3 The Discount Factor *(slide 7)* 4

1.4 Bellman Expectation Equation *(slide 8)* 4

1.5 Bellman Optimality *(slides 9–12)* 5

2 Policy Evaluation *(slides 13–15)* **6**

3 Exact Policy Evaluation *(slides 16–22)* **6**

4 Fixed-Point Iteration *(slides 23–24)* **7**

5 Contractions *(slides 25–28)* **8**

5.1 Contraction Operator *(slides 27–28)* 9

6 Iterative Policy Evaluation *(slides 29–32)* **10**

7 Iterative Policy Evaluation Theorem *(slides 33–37)* **11**

8 Iterative Policy Evaluation: Iterations *(slide 38)* **12**

9 How to Find the Optimal Policy? *(slides 39–43)* **12**

10 The Bellman Optimality Backup is a Contraction *(slides 44–47)* **13**

11 Bellman Operator for Q *(slide 48)* **14**

12 Value Iteration *(slides 49–53)* **14**

13 Policy from Value Iteration *(slides 54–65)* **15**

14 Number of Iterations of Value Iteration *(slides 66–74)* **17**

15 Another Note on Value Iteration *(slide 75)* **18**

16 Summary **19**

16.1 Comparison of the methods and results 19

16.2 Cost scaling 19

16.3 Key take-aways 19

A Supplementary: Banach's Theorem, Existence and Rate **20**

B Supplementary: Self-Check List from the Companion Note **21**

1 Recap: Sequential Decision Making and MDPs *(slides 2–12)*

This block repeats what the previous lecture (L03) established. Everything later is built on it, and three of the results are quoted from L03 without proof (Facts F1–F4 below).

1.1 Sequential Decision Making and the MDP *(slides 3–4)*

The agent interacts with its environment at discrete timesteps: at each step it receives an observation o_t and a reward r_t from the environment and answers with an action a_t (Figure 1). The *state* is a function of the history, $s_t = f(h_t)$, and is typically hidden or unknown.

A **Markov Decision Process** is sequential decision making under the Markov assumption: the transition dynamics are Markovian, the state is fully observable, and the dynamics are (in general) stochastic, $p(s_{t+1} \mid s_t, a_t)$. Equivalent shorthand: $s_{t+1} \sim p(\cdot \mid s_t, a_t)$ or $s' \sim p(\cdot \mid s, a)$. An MDP is the tuple $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$ with set of states $\mathcal{S}$, set of actions $\mathcal{A}$, transition kernel p, reward $r(s, a)$ and discount γ.

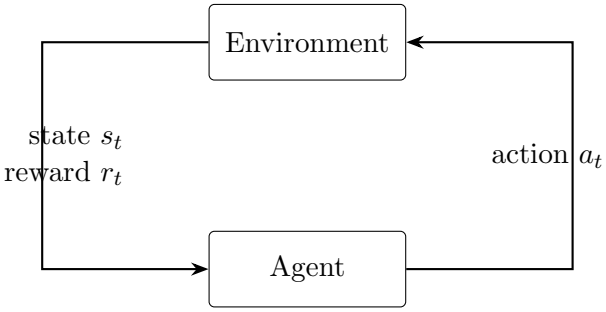


Figure 1: The agent–environment loop (redrawn from slides 3–4). The environment answers each action with a new state/observation and a reward.

Intuition

The Markov property says that *the present state is a sufficient summary of the past*: once you know s_t and a_t, the history tells you nothing more about where you will go next. That is why a policy can be a function of the current state alone, and why value functions can be defined state by state.

1.2 Policies, Value Functions and Q-Functions *(slides 5–6)*

A **policy** π is a mapping from (all) states to actions; it determines how the agent selects actions. It can be *deterministic*, $a = \pi(s)$, or *stochastic*, $\pi(a \mid s)$ (also written $p(a \mid s)$ or $a \sim \pi(\cdot \mid s)$).

The **value function** and **Q-function** of π measure how good states and actions are, and give a way to compare policies:

$$V^\pi(s) = \mathbb{E}_\pi\Big[\sum_{h\geq 0} \gamma^h r_{t+h} \;\Big|\; s_t = s\Big], \qquad Q^\pi(s,a) = \mathbb{E}_\pi\Big[\sum_{h\geq 0} \gamma^h r_{t+h} \;\Big|\; s_t = s,\, a_t = a\Big],$$

where after the first step actions follow π and next states are drawn from p. For infinite-horizon MDPs there always exists a *deterministic* policy π^* with $V^{\pi^*}(s) \geq V^\pi(s)$ for all s and all π: the optimal policy *dominates every other policy in every state*. We write $V^* = V^{\pi^*}$ and $Q^* = Q^{\pi^*}$.

Intuition

Comparing two policies means comparing two vectors $V^{\pi}, V^{\pi'} \in \mathbb{R}^{|\mathcal{S}|}$, which is only a *partial* order: π might be better in one state and worse in another. The nontrivial fact is that for discounted infinite-horizon MDPs a single deterministic π^* is best *in every state at once*, so "find the optimal policy" is a well-posed goal.

1.3 The Discount Factor *(slide 7)*

Setting $\gamma = 1$ for infinite tasks is a bad idea: the sum of rewards may diverge. For $\gamma \in [0, 1)$ the weights form a geometric series,

$$\sum_{h=0}^{\infty} \gamma^h = \frac{1}{1-\gamma},$$

so the value of γ roughly determines *how many steps ahead we are looking*: $\gamma = 0.99$ corresponds to a horizon of about 100 steps (the slide rounds this to "99 timesteps ahead"). This also bounds all values:

Fact F1 (L03-1, Lemma 3 — range of values)**.** If $r(s,a) \in [0, R_{\max}]$ then $0 \le V^{\pi}(s), V^{*}(s), Q^{\pi}(s,a), Q^{*}(s,a) \le R_{\max}/(1-\gamma)$. (Each is a discounted sum of terms in $[0, R_{\max}]$, bounded by $R_{\max} \sum_h \gamma^h$.) With $R_{\max} = 1$ this is the interval $[0, 1/(1-\gamma)]$ used on the slides.

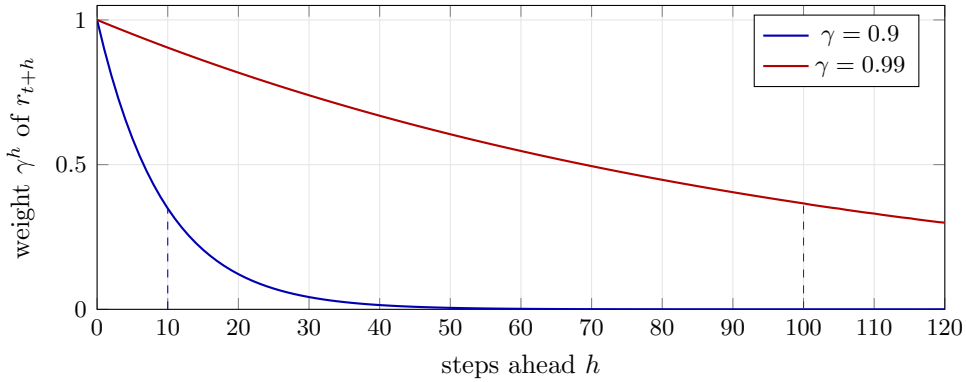


Figure 2: Weight of the reward received h steps ahead (original figure; slide 7 only describes this in prose). Dashed lines mark the effective horizon $1/(1-\gamma)$, where the weight has dropped to about $e^{-1} \approx 0.37$.

Intuition

Think of γ as a "patience" dial. With $\gamma = 0.9$ rewards more than ~ 10 steps away barely matter; with $\gamma = 0.99$ the agent looks ~ 100 steps ahead. The same number $1/(1-\gamma)$ reappears in this lecture as (i) the largest possible value, (ii) the initialisation range for the iterations, and (iii) the factor that inflates the error of the greedy policy. Larger γ means a longer horizon and harder computation.

1.4 Bellman Expectation Equation *(slide 8)*

The value of a state is expanded in terms of the current reward and the value of the next states, according to the policy:

$$V^{\pi}(s) = r(s, \pi(s)) + \gamma\, \mathbb{E}_{s' \sim p(\cdot|s,\pi(s))}\big[V^{\pi}(s')\big],$$
$$Q^{\pi}(s,a) = r(s,a) + \gamma\, \mathbb{E}_{s' \sim p(\cdot|s,a)}\big[V^{\pi}(s')\big], \qquad V^{\pi}(s) = Q^{\pi}(s, \pi(s)).$$

In the first line the reward is a function of s and $\pi(s)$; in the second, of s and a.

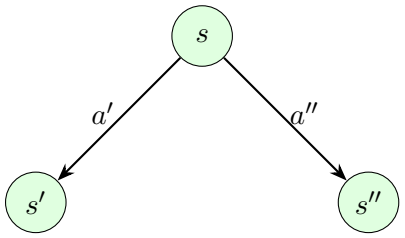
Fact F2 (L03-1, Theorem 6 — Bellman expectation equation)**.** For every policy π and every state s, the display above holds. In operator language (Definition 3): $\mathcal{B}^\pi V^\pi = V^\pi$.

Intuition

A one-step lookahead: *what you get now* plus *the discounted value of wherever you land*, averaged over the randomness of the transition. It turns an infinite sum into a *local* consistency condition that must hold in all states simultaneously, and that is exactly what we will exploit to compute V^π.

1.5 Bellman Optimality *(slides 9–12)*

Example (slide 9). Suppose we already know V^* at the successor states s' and s''. From s we try each action: taking a' gives reward $r(s,a')$ and leads to s', so $Q^*(s,a') = r(s,a') + \gamma V^*(s')$; likewise $Q^*(s,a'') = r(s,a'') + \gamma V^*(s'')$. Then $V^*(s) = \max\{Q^*(s,a'), Q^*(s,a'')\}$ (Figure 3; the slide draws deterministic transitions, with stochastic ones $V^*(s')$ becomes an expectation).



$$Q^*(s,a') = r(s,a') + \gamma V^*(s')$$
$$Q^*(s,a'') = r(s,a'') + \gamma V^*(s'')$$

$$V^*(s) = \max\{\, Q^*(s,a'),\, Q^*(s,a'') \,\}$$

Figure 3: Bellman optimality backup in one picture (redrawn from slide 9): try every action, evaluate it with the (assumed known) values of the successors, keep the best.

In general,

$$V^*(s) = \max_a \Big[r(s,a) + \gamma\, \mathbb{E}_{s' \sim p(\cdot|s,a)} V^*(s') \Big], \qquad Q^*(s,a) = r(s,a) + \gamma\, \mathbb{E}_{s' \sim p(\cdot|s,a)} \Big[\max_{a'} Q^*(s',a') \Big].$$

Two results from L03 are quoted on the slides:

Fact F3 (Slide 10 "Theorem 1", L03-1 Theorem 8 — Bellman optimality)**.** V^* and Q^* satisfy the two equations above, and the greedy policy $\hat{\pi}(s) \in \arg\max_a Q^*(s,a)$ satisfies $V^{\hat{\pi}} = V^*$. Hence $\pi^*(s) = \arg\max_a Q^*(s,a)$ is an optimal policy.

Fact F4 (Slide 11 "Theorem 2", L03-1 Theorem 10 — uniqueness)**.** For any V, if $V(s) = \max_a \big[r(s,a) + \gamma \mathbb{E}_{s' \sim p(\cdot|s,a)} V(s') \big]$ for all s, then $V = V^*$.

Intuition

The optimality equation lets us *focus on one step at a time*: we only choose the best first action and leave the rest of the problem to $V(s')$. Fact F4 says that any function that is locally self-consistent in this sense *is* V^*: there is nothing else to look for. Fact F3 converts Q^* into a policy by a plain arg max. Together they say: *find the unique solution of the Bellman optimality equation and you are done.* How to find it is the subject of the rest of the lecture. (The proof of uniqueness in L03 was a long unrolling argument; Corollary 5 below re-proves it in one line.) *(slide 12)* closes the recap.

2 Policy Evaluation *(slides 13–15)*

Question. Given an MDP $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$ and a policy π, how can we compute the goodness of π, i.e. V^π?

Why is this interesting? There are $|\mathcal{A}|^{|\mathcal{S}|}$ possible deterministic policies, and we want to find the optimal one. To find it, we need to be able to *evaluate* candidates.

Intuition

Policy evaluation is the "scoring function" of reinforcement learning: before you can say that one policy is better than another you need a number (really, a vector with one entry per state) that says how well each one does. Everything in this lecture is about computing such numbers cheaply: first for a *given* policy, then for the *best* policy at once.

3 Exact Policy Evaluation *(slides 16–22)*

Note: the companion note marks this block as "on the slides only": it is not proved there, so prepare it from the slides.

Given $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$ and π, what is V^π? We know that the Bellman expectation equation holds for *all* states:

$$V^\pi(s) = r(s, \pi(s)) + \gamma \sum_{s' \in \mathcal{S}} p(s' \mid s, \pi(s))\, V^\pi(s').$$

How many linear constraints do we have? $|\mathcal{S}|$, one per state, in the $|\mathcal{S}|$ unknowns $V^\pi(s)$. We can combine all the constraints (slide credit: Wen Sun). Define the reward vector and the transition matrix induced by π:

$$r^\pi \in \mathbb{R}^{|\mathcal{S}|},\ (r^\pi)_s = r(s, \pi(s)); \qquad P^\pi \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{S}|},\ (P^\pi)_{s,s'} = p(s' \mid s, \pi(s)).$$

Then the $|\mathcal{S}|$ equations read $V^\pi = r^\pi + \gamma P^\pi V^\pi$ (Figure 4), which we can solve for V^π.

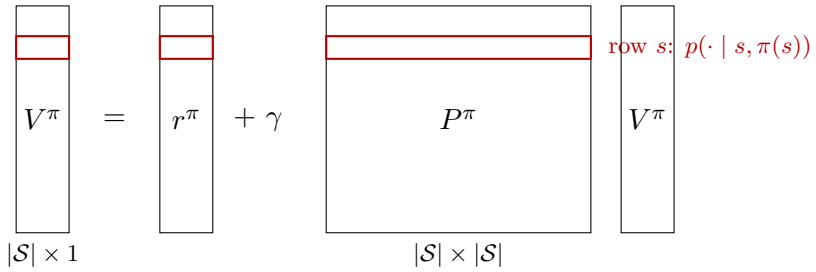


Figure 4: The $|\mathcal{S}|$ Bellman expectation equations stacked in matrix form (redrawn from slides 20–22). The highlighted row is the equation for one state s: $V^\pi(s) = r(s, \pi(s)) + \gamma \langle p(\cdot \mid s, \pi(s)), V^\pi \rangle$.

Proposition S1 (Exact policy evaluation, slides 21–22)**.** *For $\gamma < 1$, the matrix $I - \gamma P^\pi$ is invertible and*

$$V^\pi = (I - \gamma P^\pi)^{-1} r^\pi = \sum_{k \geq 0} \gamma^k (P^\pi)^k r^\pi.$$

Computing it by matrix inversion costs $O(|\mathcal{S}|^3)$.

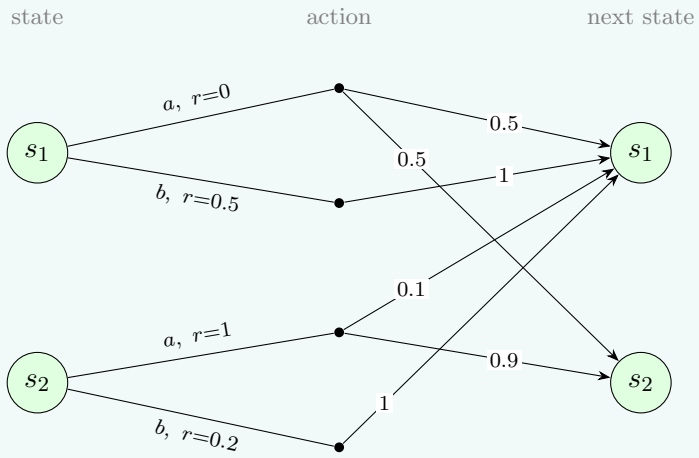
Proof. P^π is row-stochastic (entries ≥ 0, each row sums to 1), so its induced ∞-norm is 1 and $\|\gamma P^\pi x\|_\infty \leq \gamma \|x\|_\infty$. If $(I - \gamma P^\pi)x = 0$ then $x = \gamma P^\pi x$, so $\|x\|_\infty \leq \gamma \|x\|_\infty$, forcing $x = 0$ since $\gamma < 1$; hence the matrix is invertible. The Neumann series $\sum_k (\gamma P^\pi)^k$ converges (its terms have norm $\leq \gamma^k$) and equals the inverse. Rearranging $V^\pi = r^\pi + \gamma P^\pi V^\pi$ gives $(I - \gamma P^\pi)V^\pi = r^\pi$. Inverting an $|\mathcal{S}| \times |\mathcal{S}|$ matrix takes $O(|\mathcal{S}|^3)$ operations. □

Intuition

The Neumann series is the Bellman equation unrolled: $(P^\pi)^k r^\pi$ is the expected reward k steps ahead, and γ^k discounts it, so V^π is literally the *expected discounted sum of rewards*. The closed form is "nice but computationally expensive" (the slide's own verdict): cubic in the number of states is fine for dozens of states but hopeless for millions. This motivates an approach that only needs matrix–vector products.

Example E1 (A two-state MDP, used throughout)**.** Take $\mathcal{S} = \{s_1, s_2\}$, $\mathcal{A} = \{a, b\}$, $\gamma = 0.9$, with

(s, action)	r	$p(s_1 \mid \cdot)$	$p(s_2 \mid \cdot)$
(s_1, a)	0	0.5	0.5
(s_1, b)	0.5	1	0
(s_2, a)	1	0.1	0.9
(s_2, b)	0.2	1	0



Filled dots are state-action pairs; the arrows leaving a dot are the stochastic transitions, labelled with their probabilities. For the policy $\pi = (a, a)$ (always play a):

$$P^\pi = \begin{pmatrix} 0.5 & 0.5 \\ 0.1 & 0.9 \end{pmatrix}, \quad r^\pi = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad I - \gamma P^\pi = \begin{pmatrix} 0.55 & -0.45 \\ -0.09 & 0.19 \end{pmatrix}, \quad \det = 0.064,$$

$$V^\pi = (I - \gamma P^\pi)^{-1} r^\pi = \frac{1}{0.064} \begin{pmatrix} 0.19 & 0.45 \\ 0.09 & 0.55 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 7.03125 \\ 8.59375 \end{pmatrix}.$$

It will turn out that this policy is in fact optimal for this MDP: $V^* = (7.03125, 8.59375)$.

4 Fixed-Point Iteration *(slides 23–24)*

What is a **fixed point**? A point x such that $x = f(x)$. How can we find such points?

1. Initialise x_0.
2. Repeat $x_{i+1} = f(x_i)$.
3. Stop at convergence, when x does not change any more.

Geometrically (slide 24), fixed points are the intersections of $y = f(x)$ with the line $y = x$. The iteration is a "cobweb": from (x_i, x_i) on the diagonal go vertically to the curve at (x_i, x_{i+1}), then

horizontally back to the diagonal at (x_{i+1}, x_{i+1}). The slide's example is $f(x) = \cos x$, with fixed point $x^* \approx 0.739085$.

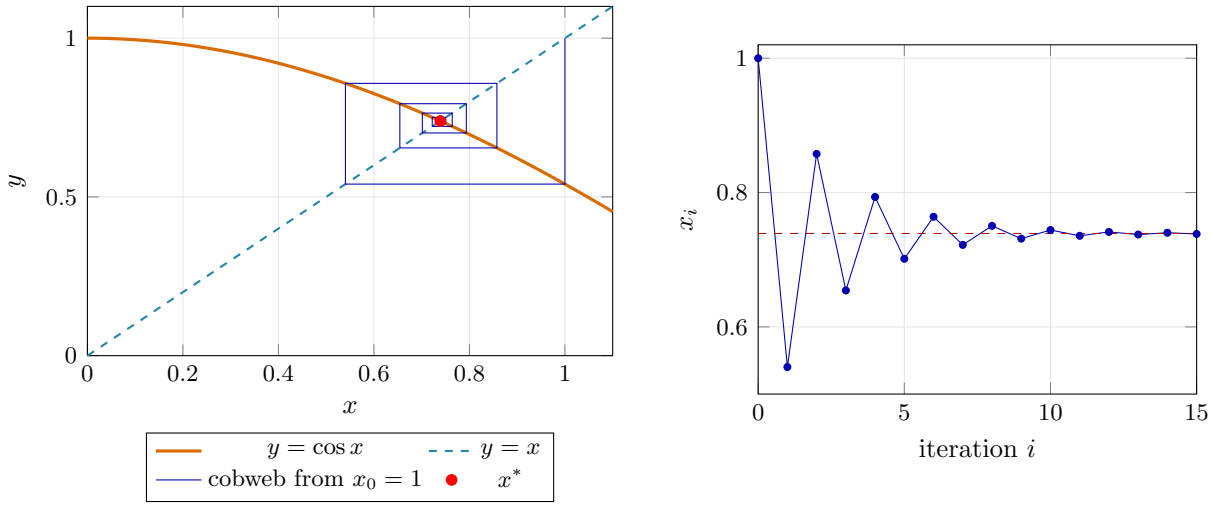


Figure 5: Fixed-point iteration of $f(x) = \cos x$ from $x_0 = 1$ (original figure in the style of slide 24; values computed exactly). Left: cobweb diagram. Right: $x_i \to x^* \approx 0.739085$.

Intuition

Fixed-point iteration is "keep applying the rule until it stops changing anything". Whether this works depends on the rule: with $f(x) = \cos x$ each step lands closer to x^* (the cobweb spirals in), but for $f(x) = 2x$ the iterates would run away from the fixed point 0. The property that guarantees spiralling *in* is that f brings any two points closer together, which is exactly what the next section calls a contraction.

5 Contractions *(slides 25–28)*

Convergence to a fixed point is possible thanks to the existence of **contraction mappings**.

Definition 1 (Contraction)**.** Let (M, d) be a metric space. An operator $f : M \to M$ is a *k-contraction* if there is $k \in [0, 1)$ with

$$d\big(f(x), f(y)\big) \le k\, d(x, y) \qquad \text{for all } x, y \in M.$$

(On the slides: $|f(x) - f(x')| \le k\,|x - x'|$ for $k \in [0, 1)$.)

Theorem 2 (Banach fixed-point theorem)**.** *A k-contraction on a non-empty* complete *metric space has* exactly one *fixed point x^*, and for any starting point x_0 the iteration $x_{i+1} = f(x_i)$ converges to it, with $d(x_i, x^*) \le k^i\, d(x_0, x^*)$.*

Proof of uniqueness (this is the part on slide 25: "a contraction mapping has at most one fixed point"). If $f(x) = x$ and $f(y) = y$ then $d(x, y) = d(f(x), f(y)) \le k\, d(x, y)$, and since $k < 1$ this forces $d(x, y) = 0$, i.e. $x = y$. □

The existence of a fixed point, the convergence of the iteration and the rate k^i are not proved on the slides; their proof is in Appendix A (supplementary).

Intuition

A contraction is a map that *squeezes space*: any two points end up closer together after one application, by a factor of at least $k < 1$. Two fixed points would stay at the same distance after applying f, which contradicts the squeezing, so there is at most one. Iterating f keeps squeezing the whole space towards a single point, which is the fixed point; completeness just guarantees that this point exists inside the space instead of being a "hole". The rate k^i is the geometric decay that all later bounds inherit.

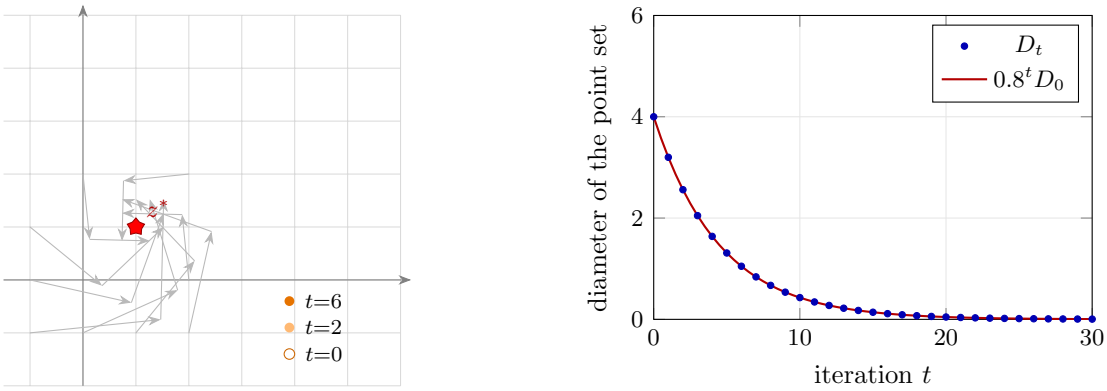


Figure 6: A contraction in $\mathbb{R}^2$ (original figure in the style of slide 26): $f(z) = 0.8\, R_{30^\circ} z + c$ with unique fixed point z^*. Left: nine starting points and their images after 2 and 6 steps all collapse onto z^*. Right: the diameter of the point set decays exactly like $k^t D_0$ (here f is a similarity, so the contraction inequality is tight).

Example E2 (Why cos is a contraction on $[0,1]$)**.** *(Added example connecting the cobweb of slide 24 to Banach's theorem; not on the slides.)* On $M = [0,1]$ (complete) cos maps into $[\cos 1, 1] \approx [0.540, 1] \subset M$, and by the mean value theorem $|\cos x - \cos y| = |\sin \xi|\ |x - y| \le \sin(1)\ |x - y|$ with $\sin 1 \approx 0.841$. So cos is a 0.841-contraction on $[0,1]$; Theorem 2 gives a unique fixed point ($x^* \approx 0.739085$) and $|x_i - x^*| \le 0.841^i\, |x_0 - x^*|$. In Figure 5, $x_0 = 1$ gives $|x_5 - x^*| \approx 0.038$, comfortably below the bound $0.841^5 \cdot 0.261 \approx 0.110$.

5.1 Contraction Operator *(slides 27–28)*

In the simplest case the contraction can be an *operator*, even as simple as a matrix O acting on value vectors:

$$\left|OV - OV'\right| \le \gamma \left|V - V'\right|$$

(we can write γ instead of k, since both lie in $[0,1)$). The slide's demo takes $O = \gamma I$, "pure shrink": every vector is scaled by γ, so the distance between OV and OV' is exactly γ times the distance between V and V' (Figure 7).

Intuition

The point of this slide is the *reading* of "contraction" for value functions: think of each value function as a point in $\mathbb{R}^{|\mathcal{S}|}$ (one coordinate per state); an operator is a contraction when it makes two different value functions *more similar*. The Bellman backups we are about to study are not as trivial as γI, but they shrink distances in the same way, by at least a factor γ in sup-norm.



Figure 7: Contraction operator $O = \gamma I$ with $\gamma = 0.6$ (original figure in the style of slide 28; Euclidean norm). Applying O shrinks the distance between any two value vectors from 2.236 to $1.342 = 0.6 \times 2.236$.

6 Iterative Policy Evaluation *(slides 29–32)*

Instead of inverting a matrix, we exploit the fixed-point structure $V^\pi = r^\pi + \gamma P^\pi V^\pi$ and *iterate* it. We assume rewards are in $[0, 1]$.

Algorithm A (Iterative policy evaluation)**.** *Input:* an MDP $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$ and a policy π.

1. Initialise $V_0 \in [0, 1/(1-\gamma)]^{|\mathcal{S}|}$ (typically $V_0 = 0$).
2. Until convergence: $V_{i+1} = r^\pi + \gamma P^\pi V_i$ (matrix form: all states are updated at once).

In operator notation, $V_{i+1} = \mathcal{B}^\pi V_i$: this is the *Bellman expectation backup*.

(In the companion note: "repeat $V_{i+1} = \mathcal{B}^\pi V_i$ until $\|V_{i+1} - V_i\|_\infty$ is small.")

Why initialise in $[0, 1/(1-\gamma)]$? Under the assumption $r(s, a) \in [0, 1]$, the maximum possible value is $1 + \gamma + \gamma^2 + \cdots = 1/(1-\gamma)$ (Fact F1). There is no reason to initialise higher than that, since no true value lives there.

Definition 3 (Bellman operators)**.** For a policy π, a value function V and a state-action function Q:

$$(\mathcal{B}^\pi V)(s) = r(s, \pi(s)) + \gamma\, \mathbb{E}_{s' \sim p(\cdot|s,\pi(s))}\big[V(s')\big] \qquad \text{(expectation backup, slide 32)},$$

$$(\mathcal{B} V)(s) = \max_a \Big[r(s, a) + \gamma\, \mathbb{E}_{s' \sim p(\cdot|s,a)}\big[V(s')\big]\Big] \qquad \text{(optimality backup, slide 44)},$$

$$(\mathcal{B} Q)(s, a) = r(s, a) + \gamma\, \mathbb{E}_{s' \sim p(\cdot|s,a)}\Big[\max_{a'} Q(s', a')\Big] \qquad \text{(on } Q\text{, slide 48)}.$$

$\mathcal{B}^\pi$ is *affine* ($\mathcal{B}^\pi V = r^\pi + \gamma P^\pi V$); $\mathcal{B}$ is *not* linear, because of the maximum. (Part of the literature writes T for $\mathcal{B}$.)

Intuition

A "backup" is one round of "look one step ahead and update". The expectation backup $\mathcal{B}^\pi$ asks: *if I act according to π for one step and then trust my current guess V, what is the value of each state?* The optimality backup $\mathcal{B}$ asks the same but lets me *pick the best action* in the first step. Facts F2 and F3 say precisely that V^π and V^* are the states of the world these backups leave untouched.

Example E3 (Iterative evaluation on the two-state MDP)**.** For $\pi = (a, a)$ in Example E1 and $V_0 = 0$ (so $\|V_0 - V^\pi\|_\infty = 8.594$):

i	$V_i(s_1)$	$V_i(s_2)$	$\|V_i - V^\pi\|_\infty$	$\gamma^i \|V_0 - V^\pi\|_\infty$
0	0.000	0.000	8.594	8.594
1	0.000	1.000	7.594	7.734
2	0.450	1.810	6.784	6.961
3	1.017	2.507	6.087	6.265
4	1.586	3.122	5.472	5.638
5	2.118	3.671	4.922	5.075
6	2.605	4.165	4.429	4.567
7	3.046	4.608	3.986	4.110
8	3.444	5.006	3.587	3.699
9	3.803	5.365	3.229	3.329
10	4.126	5.688	2.906	2.996
11	4.416	5.979	2.615	2.697
12	4.678	6.240	2.354	2.427
13	4.913	6.476	2.118	2.184
14	5.125	6.687	1.906	1.966
15	5.315	6.878	1.716	1.769

The first sweep gives $V_1 = r^\pi = (0, 1)$; each later sweep propagates reward information one step further. The true error always stays below the geometric bound (last column), as proved next.

7 Iterative Policy Evaluation Theorem *(slides 33–37)*

The slides' claim is that, for any two value functions U and V and for all $s \in \mathcal{S}$,

$$|(\mathcal{B}^\pi U)(s) - (\mathcal{B}^\pi V)(s)| \le \gamma \|U - V\|_\infty, \qquad \|V\|_\infty = \max_s |V(s)|$$

("the largest difference between state values"). The companion note packages this together with the analogous claim for $\mathcal{B}$ (proved later, slides 44–47):

Theorem 4 (Both operators are γ-contractions in the sup-norm)**.** *For all value functions U, V:*

$$\textit{(i)}\ \ \|\mathcal{B}^\pi U - \mathcal{B}^\pi V\|_\infty \le \gamma \|U - V\|_\infty, \qquad \textit{(ii)}\ \ \|\mathcal{B}U - \mathcal{B}V\|_\infty \le \gamma \|U - V\|_\infty.$$

The same holds for $\mathcal{B}$ acting on state-action functions Q.

Proof of (i), as on slides 33–37. For $\mathcal{B}^\pi$ the rewards cancel. For every state s,

$$\begin{aligned} |(\mathcal{B}^\pi U)(s) - (\mathcal{B}^\pi V)(s)| &= \gamma \left| \mathbb{E}_{s'} \big[U(s') - V(s') \big] \right| \\ &\le \gamma\, \mathbb{E}_{s'} \left| U(s') - V(s') \right| && \text{(Jensen: } |\cdot| \text{ is convex)} \\ &\le \gamma\, \|U - V\|_\infty && \text{(an average is at most the maximum).} \end{aligned}$$

The inequality holds for every s, so also for the state where the left-hand side is largest; taking the maximum over s gives (i). The proof of (ii) is in Section 10. □

Intuition

Two things happen in the proof. First, both backups use the *same* rewards, so these cancel: the only thing that differs is the next-state values, scaled by γ. Second, the next-state values are *averaged* by probabilities that sum to one, and an average of numbers each at most $\|U - V\|_\infty$ in size cannot exceed $\|U - V\|_\infty$. So no state can end up further apart than γ times the worst

gap before. As the slide remarks, we are not secretly using the sup-norm on the left-hand side: the bound holds *for each state*, including the one with the largest difference.

Slide 34 draws the conclusion: *hence the Bellman expectation backup is a contraction mapping and V^π is its fixed point.* This follows from Theorem 2 applied to $\mathcal{B}^\pi$ on $(\mathbb{R}^{|\mathcal{S}|}, \|\cdot\|_\infty)$, which is complete (a finite-dimensional normed space): by Fact F2, $\mathcal{B}^\pi V^\pi = V^\pi$, so V^π is a fixed point, and by Banach's uniqueness it is the only one. (This is part (a) of Corollary 5, stated in full in the Value Iteration section.)

8 Iterative Policy Evaluation: Iterations *(slide 38)*

Corollary 6 (Convergence and cost of iterative policy evaluation)**.** $\|V_i - V^\pi\|_\infty \le \gamma^i \|V_0 - V^\pi\|_\infty$, *so* $\|V_i - V^\pi\|_\infty \le \epsilon$ *after*

$$i \;\ge\; \frac{\ln\big(\|V_0 - V^\pi\|_\infty / \epsilon\big)}{\ln(1/\gamma)}$$

iterations, each costing $O(|\mathcal{S}|^2)$ arithmetic operations. Hence an ϵ-accurate value function costs $O\big(|\mathcal{S}|^2 \ln(1/\epsilon)\big)$ (treating γ and $\|V_0 - V^\pi\|_\infty$ as constants).

Proof. The rate is Theorem 2 with $k = \gamma$, applied to $\mathcal{B}^\pi$ whose unique fixed point is V^π. Solving $\gamma^i \|V_0 - V^\pi\|_\infty \le \epsilon$ for i (take logarithms; $\ln \gamma = -\ln(1/\gamma) < 0$ flips the inequality) gives the count. Each sweep updates $|\mathcal{S}|$ states and each update averages over $|\mathcal{S}|$ successors, i.e. one matrix–vector product $P^\pi V_i$. □

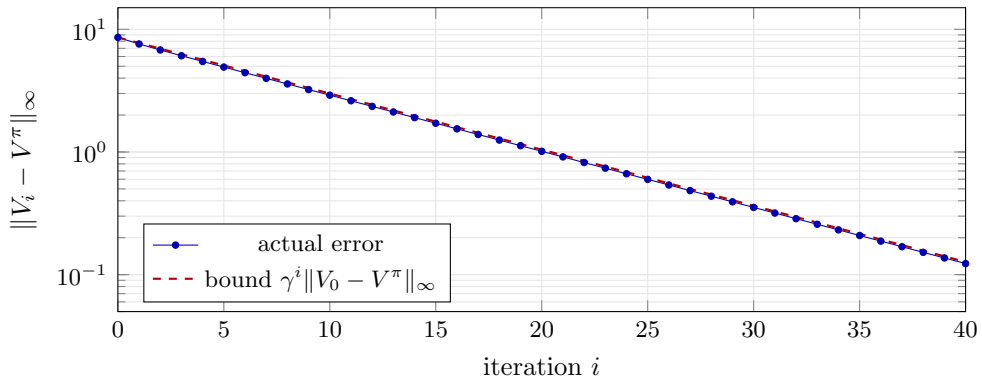


Figure 8: Iterative policy evaluation on Example E1 with $\pi = (a, a)$, $\gamma = 0.9$, $V_0 = 0$. On a log scale the error is a straight line: geometric convergence. The error decays geometrically and remains below the contraction bound.

Intuition

Worked numbers. With $\gamma = 0.9$, $\epsilon = 0.01$ and $\|V_0 - V^\pi\|_\infty = 8.594$: $i \ge \ln(859.4)/\ln(1/0.9) = 6.756/0.1054 = 64.1$, so 65 sweeps (indeed $0.9^{65} \cdot 8.594 \approx 0.009$). Each sweep is only $|\mathcal{S}|^2 = 4$ multiplications here. The *accuracy* costs only a logarithm ($10\times$ smaller ϵ adds $\ln 10/\ln(1/\gamma) \approx 22$ sweeps), while the *horizon* $\ln(1/\gamma) \approx 1 - \gamma$ drives the cost. Compared with $O(|\mathcal{S}|^3)$ for exact evaluation, iteration wins when $|\mathcal{S}|$ is large and ϵ is moderate.

9 How to Find the Optimal Policy? *(slides 39–43)*

Now what we are really interested in is finding the optimal policy π^*.

Naive approach (slides 40–42): we know how to do policy evaluation, so

- for each possible policy, for all states,
- do policy evaluation and compute $V^{\pi}(s)$,
- choose π' such that $V^{\pi'}(s) \geq V^{\pi}(s)$ (keep the best).

If we use exact policy evaluation the cost is $O\big(|\mathcal{A}|^{|\mathcal{S}|}\,|\mathcal{S}|^3\big)$: $|\mathcal{A}|^{|\mathcal{S}|}$ policies, each needing a cubic solve.

Intuition

The slide's horrified emoji is justified. The number of deterministic policies grows *exponentially* in the number of states: for $|\mathcal{A}| = 2$, $|\mathcal{S}| = 10$ there are 1024 policies (about 10^6 operations in total), but for $|\mathcal{S}| = 30$ there are $2^{30} \approx 10^9$ policies and about 2.9×10^{13} operations. Enumeration treats the policies as unrelated black boxes. The better idea (slide 43): **let's use the Bellman optimality equation!** Rather than searching over policies, search for the one function V^* (or Q^*) that satisfies it, and read the policy off with an arg max (Fact F3).

10 The Bellman Optimality Backup is a Contraction *(slides 44–47)*

Set $\gamma < 1$ and use the sup-norm $\|V\|_\infty = \max_s |V(s)|$. Define the (non-linear) operator $\mathcal{B}V$ as the Bellman optimality equation applied to V (Definition 3):

$$(\mathcal{B}V)(s) = \max_a \Big[r(s,a) + \gamma \sum_{s' \in \mathcal{S}} p(s' \mid s,a)\, V(s') \Big].$$

We need one elementary fact about maxima, used implicitly on the slides:

Lemma L1 (The maximum is 1-Lipschitz)**.** *For any two functions f, g on a finite set of actions, $\big|\max_a f(a) - \max_a g(a)\big| \leq \max_a |f(a) - g(a)|$.*

Proof. Let a_f attain $\max_a f$. Then $\max_a f - \max_a g \leq f(a_f) - g(a_f) \leq \max_a |f(a) - g(a)|$. Exchanging the roles of f and g gives the other sign. □

Proof of Theorem 4 (ii), following slides 45–47. Fix two value functions V_k, V_j and a state s. Using Lemma L1 with $f(a) = r(s,a) + \gamma \sum_{s'} p(s' \mid s,a) V_k(s')$ and g the same with V_j, the reward terms cancel:

$$\begin{aligned} \big|(\mathcal{B}V_k)(s) - (\mathcal{B}V_j)(s)\big| &\leq \max_a \Big| \gamma \sum_{s'} p(s' \mid s,a) \big(V_k(s') - V_j(s')\big) \Big| \\ &\leq \max_a \gamma \sum_{s'} p(s' \mid s,a)\, \|V_k - V_j\|_\infty \\ &= \gamma \|V_k - V_j\|_\infty, \end{aligned}$$

where the second step bounds each $|V_k(s') - V_j(s')|$ by the sup-norm, and the last equality uses $\sum_{s'} p(s' \mid s,a) = 1$. Taking the maximum over s gives $\|\mathcal{B}V_k - \mathcal{B}V_j\|_\infty \leq \gamma \|V_k - V_j\|_\infty$. □

Intuition

If you apply $\mathcal{B}$ to two different value functions, *the distance between them shrinks* (slide 46): $\mathcal{B}$ makes value functions closer by at least a factor γ (slide 47). The new ingredient compared with the expectation backup is the max, and Lemma L1 says it cannot amplify differences:

changing a few action values by at most δ changes the best of them by at most δ. So even though $\mathcal{B}$ is non-linear, it inherits the same squeezing property as $\mathcal{B}^\pi$.

11 Bellman Operator for Q *(slide 48)*

Since $Q : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$, also $\mathcal{B}Q : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$:

$$(\mathcal{B}Q)(s,a) = r(s,a) + \gamma\, \mathbb{E}_{s' \sim p(\cdot|s,a)} \Big[\max_{a'} Q(s',a') \Big],$$

and the norm is now $\|Q\|_\infty = \max_{s,a} |Q(s,a)|$.

Proof of the last claim of Theorem 4 (the Q-version). Rewards cancel, and the maximum now sits *inside* the expectation. For every (s,a),

$$\big|(\mathcal{B}Q)(s,a) - (\mathcal{B}Q')(s,a)\big| = \gamma \Big| \mathbb{E}_{s'} \big[\max_{a'} Q(s',a') - \max_{a'} Q'(s',a') \big] \Big| \le \gamma\, \mathbb{E}_{s'} \max_{a'} \big| Q(s',a') - Q'(s',a') \big| \le \gamma \|Q - Q'\|_\infty$$

using $|\mathbb{E}X| \le \mathbb{E}\,|X|$, Lemma L1, and that an average is at most the maximum. Taking the maximum over (s,a) finishes the proof. □

Intuition

Nothing new happens: the Q-table has $|\mathcal{S}||\mathcal{A}|$ entries instead of $|\mathcal{S}|$, and the same "rewards cancel, averages cannot exceed the worst case" argument applies. Working with Q is convenient because a policy can be read off from Q without a model: $\arg\max_a Q(s,a)$.

12 Value Iteration *(slides 49–53)*

We can obtain $Q^* = \mathcal{B}Q^*$, since Q^* is a fixed point of $Q = \mathcal{B}Q$ (and all of this also holds for V^*).

Corollary 5 (What the fixed points are)**.** (a) *V^π is the unique fixed point of $\mathcal{B}^\pi$;* (b) *V^* is the unique fixed point of $\mathcal{B}$ on value functions;* (c) *Q^* is the unique fixed point of $\mathcal{B}$ on state-action functions.*

Proof. Fact F2 says $\mathcal{B}^\pi V^\pi = V^\pi$ and Fact F3 says $\mathcal{B}V^* = V^*$ and $\mathcal{B}Q^* = Q^*$, so each is a fixed point. The spaces $(\mathbb{R}^{|\mathcal{S}|}, \|\cdot\|_\infty)$ and $(\mathbb{R}^{|\mathcal{S}||\mathcal{A}|}, \|\cdot\|_\infty)$ are complete, and Theorem 4 says the operators are γ-contractions, so Theorem 2 says each has only one fixed point. □

(This also re-proves Fact F4, slide 11, in one line: any V with $\mathcal{B}V = V$ is the fixed point of $\mathcal{B}$, hence equals V^*; this is what the long unrolling argument in L03 did by hand.)

Algorithm B (Value iteration)**.** *Input:* an MDP $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$.

1. Initialise Q_0 with entries in $[0, 1/(1-\gamma)]$ (typically $Q_0 = 0$).
2. Until convergence, for all states and actions:
$$Q_{i+1}(s,a) = (\mathcal{B}Q_i)(s,a) = r(s,a) + \gamma \sum_{s'} p(s' \mid s,a) \max_{a'} Q_i(s',a').$$
3. Return the greedy policy $\pi_i(s) \in \arg\max_a Q_i(s,a)$ (slide 54).

Theorem 7 (Value iteration converges geometrically). $\|Q_i - Q^*\|_\infty \le \gamma^i \|Q_0 - Q^*\|_\infty$, *and each iteration costs* $O(|\mathcal{S}|^2|\mathcal{A}|)$.

Proof. By Theorem 4 and Corollary 5(c), $\|Q_{i+1} - Q^*\|_\infty = \|\mathcal{B}Q_i - \mathcal{B}Q^*\|_\infty \le \gamma \|Q_i - Q^*\|_\infty$, and induction gives $\|Q_i - Q^*\|_\infty \le \gamma^i \|Q_0 - Q^*\|_\infty$ (slide 51). For the cost (slide 53): compute $m_i(s') = \max_{a'} Q_i(s', a')$ once per state ($|\mathcal{S}||\mathcal{A}|$ operations); then each of the $|\mathcal{S}||\mathcal{A}|$ entries needs a sum over $|\mathcal{S}|$ successors, using the shared m_i. Total $O(|\mathcal{S}|^2|\mathcal{A}|)$. □

Intuition

Value iteration is the same fixed-point iteration as in the cobweb picture, run on the *vector* Q with the operator $\mathcal{B}$: start from any guess, repeatedly "look one step ahead and take the best action", and the guess is pulled towards Q^* at geometric speed. No search over policies is needed: compare $O(|\mathcal{S}|^2|\mathcal{A}|)$ per iteration, for about $\ln(1/\epsilon)/(1-\gamma)$ iterations, with the $|\mathcal{A}|^{|\mathcal{S}|}$ of the naive approach. A useful way to see how it works: after i iterations, $Q_i(s,a)$ is the best total discounted reward you can collect in the first i steps (from a terminal value of Q_0), so reward information spreads one step per iteration.

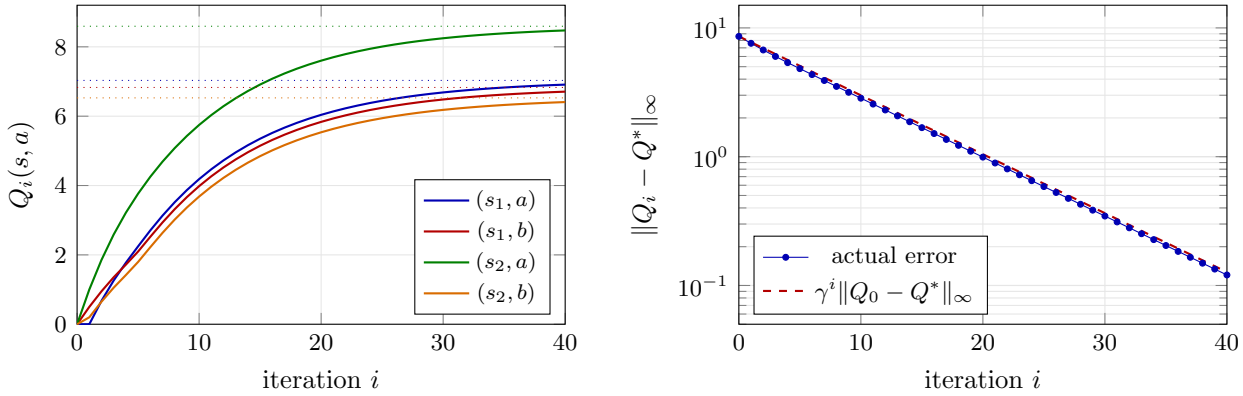


Figure 9: Value iteration on Example E1 ($\gamma = 0.9$, $Q_0 = 0$), in the spirit of the demo on slide 52. Left: the four Q-values converge to Q^* (dotted lines). Right: the error stays below the geometric bound of Theorem 7.

Example E4 (First iterations of value iteration). $Q_1 = r$ (from $Q_0 = 0$). Then, e.g., $Q_2(s_1,a) = 0 + 0.9\,(0.5 \cdot \max Q_1(s_1,\cdot) + 0.5 \cdot \max Q_1(s_2,\cdot)) = 0.9(0.5 \cdot 0.5 + 0.5 \cdot 1) = 0.675$, matching the table in Example E5. The fixed point is $Q^* = \begin{pmatrix} 7.03125 & 6.828125 \\ 8.59375 & 6.528125 \end{pmatrix}$ (rows s_1, s_2; columns a, b), so $V^* = (7.03125,\ 8.59375)$ and $\pi^* = (a, a)$.

13 Policy from Value Iteration *(slides 54–65)*

We know that $\pi^*(s) = \arg\max_a Q^*(s,a)$, and since $Q_i(s,a) \approx Q^*(s,a)$ we can choose

$$\pi_i(s) = \arg\max_a Q_i(s,a).$$

What is the quality of such a policy? For all states,

Theorem 8 (Quality of the greedy policy). *Let* $\pi_i(s) \in \arg\max_a Q_i(s,a)$. *Then for every state* s,

$$V^{\pi_i}(s) \;\ge\; V^*(s) - \frac{2\|Q_i - Q^*\|_\infty}{1-\gamma} \;\ge\; V^*(s) - \frac{2\gamma^i \|Q_0 - Q^*\|_\infty}{1-\gamma}.$$

Proof. Write $\varepsilon = \|Q_i - Q^*\|_\infty$ and fix s. Using $V^{\pi_i}(s) = Q^{\pi_i}(s, \pi_i(s))$ and $V^*(s) = Q^*(s, \pi^*(s))$ (slide 56), add and subtract $Q^*(s, \pi_i(s))$ (slide 57):

$$V^{\pi_i}(s) - V^*(s) = \underbrace{\big[Q^{\pi_i}(s, \pi_i(s)) - Q^*(s, \pi_i(s))\big]}_{\text{(I)}} + \underbrace{\big[Q^*(s, \pi_i(s)) - Q^*(s, \pi^*(s))\big]}_{\text{(II)}}.$$

Term (I). The two Q's share the same reward term, which cancels (slide 58), so

$$\text{(I)} = \gamma\, \mathbb{E}_{s' \sim p(\cdot|s, \pi_i(s))} \big[V^{\pi_i}(s') - V^*(s')\big].$$

Term (II). Insert Q_i twice (slides 59–62):

$$\text{(II)} = \underbrace{\big[Q^*(s, \pi_i(s)) - Q_i(s, \pi_i(s))\big]}_{\geq -\varepsilon} + \underbrace{\big[Q_i(s, \pi_i(s)) - Q_i(s, \pi^*(s))\big]}_{\geq 0} + \underbrace{\big[Q_i(s, \pi^*(s)) - Q^*(s, \pi^*(s))\big]}_{\geq -\varepsilon} \geq -2\varepsilon.$$

The middle bracket is non-negative because π_i maximises Q_i; the outer two are each at least $-\varepsilon$ by definition of ε.

Putting it together. Let $\delta = \min_s \big[V^{\pi_i}(s) - V^*(s)\big] \leq 0$ (no policy beats V^*). Taking the state attaining the minimum, and bounding the expectation in (I) below by δ,

$$\delta \geq \gamma\delta - 2\varepsilon \implies \delta(1-\gamma) \geq -2\varepsilon \implies \delta \geq -\frac{2\varepsilon}{1-\gamma}.$$

This is the first inequality (it holds for every s since δ is the minimum over s). The second inequality is Theorem 7. □

Remark (The slides' "repeat" version, slides 63–65). The slides stop at $V^{\pi_i}(s) - V^*(s) \geq \gamma\, \mathbb{E}_{s'}[V^{\pi_i}(s') - V^*(s')] - 2\gamma^i \|Q_0 - Q^*\|_\infty$, using $\|Q_i - Q^*\|_\infty \leq \gamma^i \|Q_0 - Q^*\|_\infty$ twice, and then say "repeat and get $1/(1-\gamma)$": plugging the same bound into the expectation on the right again and again produces the geometric series $2\gamma^i \|Q_0 - Q^*\|_\infty\, (1 + \gamma + \gamma^2 + \cdots) = 2\gamma^i \|Q_0 - Q^*\|_\infty / (1-\gamma)$. The δ-trick above is the same argument written in one line.

Intuition

Why a factor 2? The greedy choice compares actions using Q_i instead of Q^*; each of the two actions being compared (the one we pick and the true best) can be mis-valued by up to ε, in opposite directions, so the *one-step* regret is up to 2ε. *Why* $1/(1-\gamma)$*?* A one-step mistake is not a one-time cost: the policy keeps following the same imperfect scores in all later states, so mistakes accumulate over the whole effective horizon. Note the guarantee is about the *policy actually executed*, V^{π_i}, not about Q_i.

Example E5 (The greedy policy on the two-state MDP)**.** Greedy policies $\pi_i = (\pi_i(s_1), \pi_i(s_2))$ from value iteration, with $\|Q_0 - Q^*\|_\infty = 8.594$, so the bound of Theorem 8 is $\frac{2\gamma^i}{1-\gamma} \cdot 8.594 = 171.9\, \gamma^i$:

i	$Q_i(s_1,a)$	$Q_i(s_1,b)$	$Q_i(s_2,a)$	$Q_i(s_2,b)$	$\|Q_i - Q^*\|_\infty$	π_i	$\max_s(V^* - V^{\pi_i})$	bound
0	0.000	0.000	0.000	0.000	8.594	(a,a)	0.000	171.875
1	0.000	0.500	1.000	0.200	7.594	(b,a)	2.031	154.688
2	0.675	0.950	1.855	0.650	6.739	(b,a)	2.031	139.219
3	1.262	1.355	2.588	1.055	6.006	(b,a)	2.031	125.297
4	1.774	1.720	3.218	1.419	5.375	(a,a)	0.000	112.767
5	2.247	2.097	3.766	1.797	4.827	(a,a)	0.000	101.490
6	2.706	2.522	4.253	2.222	4.341	(a,a)	0.000	91.341
8	3.519	3.318	5.080	3.018	3.514	(a,a)	0.000	73.987
10	4.185	3.982	5.747	3.682	2.846	(a,a)	0.000	59.929
15	5.351	5.147	6.913	4.847	1.681	(a,a)	0.000	35.388
20	6.039	5.836	7.601	5.536	0.992	(a,a)	0.000	20.896
30	6.685	6.482	8.248	6.182	0.346	(a,a)	0.000	7.286
40	6.911	6.707	8.473	6.407	0.121	(a,a)	0.000	2.540

At $i = 0$ all Q-values tie. For $i = 1, 2, 3$ the greedy policy is (b, a) and loses 2.031 in s_1 (it takes the immediate reward 0.5 of b instead of the long-run better a); from $i = 4$ on, it is $(a, a) = \pi^*$ with zero loss.

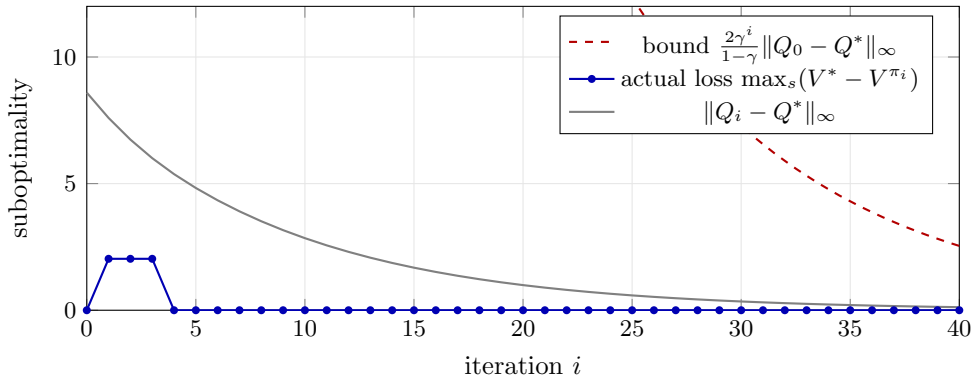


Figure 10: Guaranteed versus actual loss of the greedy policy (Example E5; original figure). The bound is vacuous (above the plotted range) until $i \approx 26$, yet the greedy policy is already optimal from $i = 4$. The theorem is a worst-case guarantee, not a prediction.

14 Number of Iterations of Value Iteration *(slides 66–74)*

Note: as with exact policy evaluation, the companion note lists this derivation as "on the slides only".

If we want an ϵ error on the quality of the policy, we can determine the number of iterations i by solving for it in

$$\frac{2\gamma^i}{1-\gamma}\|Q_0 - Q^*\|_\infty \le \epsilon.$$

Proposition S2 (Number of iterations for an ϵ-optimal greedy policy)**.** *Let $Q_0 \in [0, 1/(1-\gamma)]^{|\mathcal{S}||\mathcal{A}|}$ (e.g. $Q_0 = 0$) and rewards in $[0, 1]$. If*

$$i \;\ge\; \frac{1}{1-\gamma}\ln\Big(\frac{2}{\epsilon(1-\gamma)^2}\Big),$$

then $V^{\pi_i}(s) \ge V^*(s) - \epsilon$ for every state s.

Proof. Step 1 (slides 67–69). Q^* also lies in $[0, 1/(1-\gamma)]$ (Fact F1), so by the infinity norm $\|Q_0 - Q^*\|_\infty \le 1/(1-\gamma)$ (for $Q_0 = 0$ it is just $\|Q^*\|_\infty \le 1/(1-\gamma)$). Hence by Theorem 8 it suffices that $\dfrac{2\gamma^i}{(1-\gamma)^2} \le \epsilon$.

Step 2 (slides 70–71). Write $\gamma = 1 - (1-\gamma)$ ("add and subtract 1") and use $1 + x \le e^x$ for all real x with $x = -(1-\gamma)$:

$$\gamma^i = \big(1 - (1-\gamma)\big)^i \le e^{-(1-\gamma)i}.$$

So it suffices that $\dfrac{2e^{-(1-\gamma)i}}{(1-\gamma)^2} \le \epsilon$, i.e. $e^{-(1-\gamma)i} \le \dfrac{\epsilon(1-\gamma)^2}{2}$.

Step 3 (slides 72–74). Take logarithms and use $\ln(1/x) = -\ln x$: $-(1-\gamma)i \le -\ln\!\big(\frac{2}{\epsilon(1-\gamma)^2}\big)$. Isolating i gives the claimed condition. □

Intuition

Worked numbers ($\epsilon = 0.01$): for $\gamma = 0.9$ the condition is $i \ge 10\ln(20\,000) = 99.03$, so **100 iterations**; for $\gamma = 0.99$ it is $i \ge 100\ln(2 \cdot 10^6) = 1450.9$, so **1451 iterations**. Making γ closer to one multiplies the work by about $10\times$ per extra "nine", while making ϵ ten times smaller adds only $\ln 10/(1-\gamma) \approx 23$ iterations at $\gamma = 0.9$. In words: accuracy is cheap (logarithmic in $1/\epsilon$), the planning horizon $1/(1-\gamma)$ is what costs. Multiplying by the per-iteration cost, an ϵ-optimal policy takes $O\big(|\mathcal{S}|^2|\mathcal{A}| \cdot \frac{1}{1-\gamma} \ln \frac{1}{\epsilon(1-\gamma)^2}\big)$ arithmetic operations. (Replacing $\ln(1/\gamma)$ by its lower bound $1-\gamma$ is a safe simplification: it can only make the required i larger, e.g. 99.0 instead of 94.0 for $\gamma = 0.9$.)

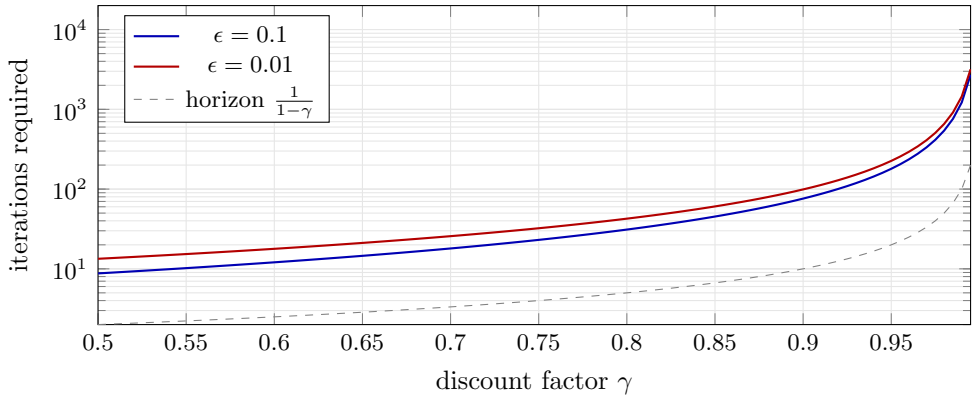


Figure 11: Iterations sufficient for an ϵ-optimal greedy policy (Proposition S2; original figure, slides 66–74 only give the formula). The curves track the horizon $1/(1-\gamma)$ up to a logarithmic factor.

15 Another Note on Value Iteration *(slide 75)*

Q_t is approximating Q^*, and from Q_t we compute a policy π_t. *However*, Q_t is generally different from Q^{π_t} until we converge to (approximately) Q^*. For example, Q_0 is just a random initial guess, which may not correspond to any policy's Q-value.

Intuition

Value iteration is *not* "evaluate a policy, then improve it". The intermediate tables Q_i are only approximations of Q^*; there may not exist any policy whose Q-function equals Q_i. In the two-state MDP, $Q_1 = r$, but every policy has strictly positive values, so $Q^\pi(s,a) = r(s,a) + \gamma \mathbb{E} V^\pi(s') > r(s,a)$ for all π: Q_1 is the Q-function of *no* policy. Only in the limit does Q_i become $Q^* = Q^{\pi^*}$. This is why Theorem 8 is needed at all: it tells us what the greedy

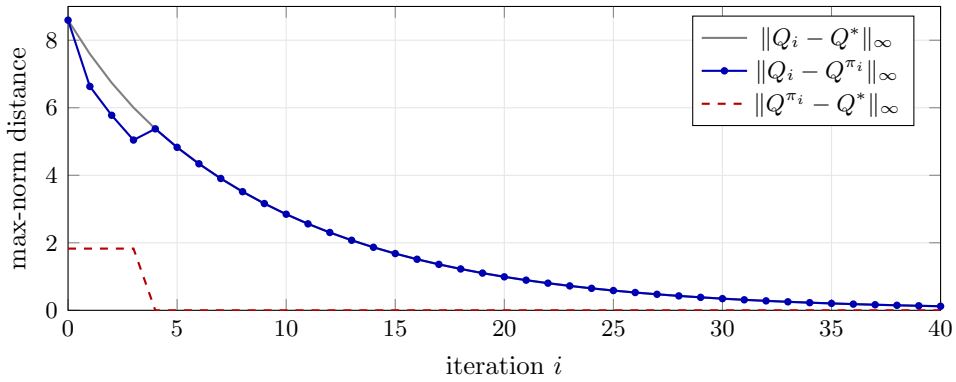


Figure 12: On Example E1: the iterate Q_i is far from the Q-function Q^{π_i} of its own greedy policy for many iterations, even after π_i has become optimal ($i \geq 4$, where $\|Q^{\pi_i} - Q^*\|_\infty = 0$). Ties at $i = 0$ are broken towards a. (Original figure; slide 75 is text only.)

policy of an imperfect Q_i is worth.

16 Summary

16.1 Comparison of the methods and results

Method	Computes / operator	Cost	Guarantee	Slides
Exact policy evaluation	V^π; solve $(I - \gamma P^\pi)V = r^\pi$	$O(\|\mathcal{S}\|^3)$	exact (Prop. S1)	16–22
Iterative policy evaluation	V^π; $\mathcal{B}^\pi$ (affine, γ-contraction)	$O(\|\mathcal{S}\|^2)$ per sweep; $O(\|\mathcal{S}\|^2 \ln \frac{1}{\epsilon})$ in total	$\|V_i - V^\pi\|_\infty \leq \gamma^i \|V_0 - V^\pi\|_\infty$ (Cor. 6)	29–38
Naive policy search	π^*; evaluate every policy	$O(\|\mathcal{A}\|^{\|\mathcal{S}\|}\|\mathcal{S}\|^3)$	exact, but exponential	39–42
Value iteration	Q^*; $\mathcal{B}$ (non-linear, γ-contraction)	$O(\|\mathcal{S}\|^2\|\mathcal{A}\|)$ per iteration	$\|Q_i - Q^*\|_\infty \leq \gamma^i \|Q_0 - Q^*\|_\infty$ (Thm. 7)	49–53
Greedy policy from VI	$\pi_i = \arg\max_a Q_i$	free (an arg max)	$V^{\pi_i} \geq V^* - \frac{2\gamma^i}{1-\gamma}\|Q_0 - Q^*\|_\infty$ (Thm. 8)	54–65
ϵ-optimal policy	number of iterations	$i \geq \frac{1}{1-\gamma} \ln \frac{2}{\epsilon(1-\gamma)^2}$	$V^{\pi_i} \geq V^* - \epsilon$ (Prop. S2)	66–74

16.2 Cost scaling

16.3 Key take-aways

- **Policy evaluation is a linear problem.** $V^\pi = (I - \gamma P^\pi)^{-1} r^\pi$ exactly in $O(|\mathcal{S}|^3)$, or approximately by iterating $V_{i+1} = r^\pi + \gamma P^\pi V_i$ at $O(|\mathcal{S}|^2)$ per sweep and $O(\ln \frac{1}{\epsilon})$ sweeps.
- **Contractions are the engine.** A k-contraction on a complete metric space has a unique fixed point and fixed-point iteration converges to it at rate k^i (Banach). Both $\mathcal{B}^\pi$ and $\mathcal{B}$ are γ-contractions in the sup-norm, because rewards cancel and averages cannot exceed maxima; hence V^π, V^* and Q^* are their unique fixed points.
- **Value iteration** $Q_{i+1} = \mathcal{B}Q_i$ converges geometrically, $\|Q_i - Q^*\|_\infty \leq \gamma^i \|Q_0 - Q^*\|_\infty$, at $O(|\mathcal{S}|^2|\mathcal{A}|)$ per iteration, avoiding the $|\mathcal{A}|^{|\mathcal{S}|}$ policy enumeration.

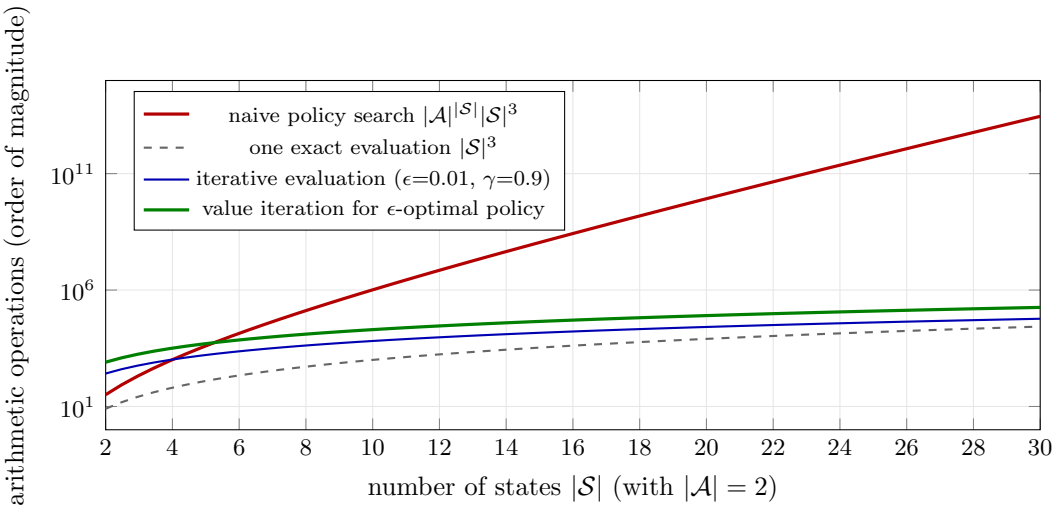


Figure 13: Qualitative cost comparison (original figure; constants dropped, iteration counts from the worked examples: 65 sweeps for evaluation, 100 iterations $\times\, |\mathcal{A}| = 2$ for value iteration, $\gamma = 0.9$, $\epsilon = 0.01$). Enumerating policies explodes exponentially; value iteration stays polynomial.

- **The greedy policy is near-optimal, with a $\frac{2}{1-\gamma}$ penalty:** $V^{\pi_i} \geq V^* - \frac{2\gamma^i}{1-\gamma}\|Q_0 - Q^*\|_\infty$, so $i \geq \frac{1}{1-\gamma} \ln \frac{2}{\epsilon(1-\gamma)^2}$ iterations give an ϵ-optimal policy: accuracy is logarithmic, the horizon $\frac{1}{1-\gamma}$ is what costs.
- **The intermediate Q_i are not policy values.** Value iteration does not alternate evaluation and improvement; Q_i is generally different from Q^{π_i} until convergence, and the bound in Theorem 8 is a worst-case guarantee that is often very pessimistic (Figure 10).

A Supplementary: Banach's Theorem, Existence and Rate

*This appendix is **not covered on the slides**; it comes from the companion note only. The slides state the uniqueness half of Theorem 2 (slide 25) and then use convergence of fixed-point iteration (slides 23–24, 29–38, 49–53) without proving it.*

Proof of Theorem 2: existence and rate. Existence. From $x_{i+1} = f(x_i)$ we get $d(x_{i+1}, x_i) = d(f(x_i), f(x_{i-1})) \leq k\, d(x_i, x_{i-1})$, hence $d(x_{i+1}, x_i) \leq k^i\, d(x_1, x_0)$ by induction. For $j > i$ the triangle inequality gives

$$d(x_j, x_i) \leq \sum_{l=i}^{j-1} d(x_{l+1}, x_l) \leq \sum_{l=i}^{j-1} k^l\, d(x_1, x_0) \leq \frac{k^i}{1-k}\, d(x_1, x_0),$$

which tends to 0 as $i \to \infty$: the sequence is Cauchy, and by completeness it has a limit x^*. A contraction is continuous (it is Lipschitz), so $f(x^*) = f(\lim x_i) = \lim f(x_i) = \lim x_{i+1} = x^*$.

Rate. Since $f(x^*) = x^*$, $d(x_i, x^*) = d(f(x_{i-1}), f(x^*)) \leq k\, d(x_{i-1}, x^*)$, and induction gives $d(x_i, x^*) \leq k^i\, d(x_0, x^*)$. □

Intuition

The sequence of iterates is a walk whose steps shrink geometrically ($d(x_{i+1}, x_i) \leq k^i d(x_1, x_0)$), so the *total remaining path length* is a convergent geometric series. A walk with summable steps cannot wander forever: it settles somewhere (completeness says that "somewhere" is a point of the space), and by continuity that limit must be unmoved by f. The rate is then just the contraction inequality applied to the pair (x_{i-1}, x^*).

B Supplementary: Self-Check List from the Companion Note

Study aid reproduced from the companion note (not a slide topic). "State" means: write the statement with assumptions and say in one sentence what it claims (for an algorithm: write its steps). "Prove" means: write the proof without looking. The last column gives the proof video of the same lecture on Classroom and where the item lives in these notes.

Item	**state**	**prove**	**Video / location here**
On the slides only: exact policy evaluation (slides 16–22)	☐	☐	Prop. S1, §3
Definition 1 (contraction)	☐		§5
Theorem 2 (Banach)	☐	☐	`ch05-banach.mp4`; §5, App. A
Definition 3 (Bellman operators)	☐		§6
Theorem 4 (both γ-contractions)	☐	☐	`ch05-contraction.mp4`; §7, §10
Corollary 5 (what the fixed points are)	☐	☐	`ch05-vi.mp4`; §12
Algorithm A (iterative policy evaluation)	☐		§6
Corollary 6 (convergence and cost)	☐	☐	`ch05-vi.mp4`; §8
Algorithm B (value iteration)	☐		§12
Theorem 7 (VI converges geometrically)	☐	☐	`ch05-vi.mp4`; §12
Theorem 8 (quality of the greedy policy)	☐	☐	`ch05-greedyquality.mp4`; §13
On the slides only: iterations of VI for an ϵ-optimal policy (slides 66–74)	☐	☐	Prop. S2, §14